

Lesson 10: Bootstrap Confidence Intervals

BSTA 551: Statistical Inference

Jessica Minnier

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Lesson 10: Bootstrap Confidence Intervals

Review: Bootstrap Basics (Lesson 9)

! The Bootstrap Principle

1. The sample represents the population $\hat{\mu}$ "resample from the sample"
2. Sampling with replacement creates variability
3. Bootstrap distribution approximates sampling distribution

Review: Bootstrap Basics (Lesson 9)

! The Bootstrap Principle

1. The sample represents the population $\hat{\theta}$ ” resample from the sample
2. Sampling with replacement creates variability
3. Bootstrap distribution approximates sampling distribution

Key algorithm:

1. Resample n observations **with replacement** from the original sample
2. Compute statistic $\hat{\theta}^*$ from bootstrap sample
3. Repeat B times (typically $B = 10,000$)
4. Analyze the distribution of $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$

Quick Review: What Bootstrap Can Estimate

Can estimate:

- Standard error of $\hat{\theta}$
- Bias of $\hat{\theta}$
- Shape/skewness of sampling distribution
- Confidence intervals

Cannot improve:

- Point estimates (centered at $\hat{\theta}$, not θ)
- A bad/unrepresentative sample

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Cannot improve:

- Point estimates (centered at $\hat{\theta}$, not θ)
- A bad/unrepresentative sample

Today's Goals:

1. Compute bootstrap standard errors
2. Construct bootstrap confidence intervals (percentile and t methods)
3. Compare methods and know when to use each
4. Apply bootstrap to complex statistics (trimmed means, ratios)

Running Example: Drug Response Times

```
1 response_times <- c(8, 12, 15, 7, 22, 45, 11, 9, 14, 18, 10, 13)
2 n <- length(response_times)
3
4 cat("n =", n, "\n")
```

n = 12

```
1 cat("Mean:", round(mean(response_times), 2), "days\n")
```

Mean: 15.33 days

```
1 cat("Median:", median(response_times), "days\n")
```

Median: 12.5 days

```
1 cat("SD:", round(sd(response_times), 2), "days")
```

SD: 10.27 days

Note the outlier at 45 days â€” this motivates robust methods!

Bootstrap Standard Errors

Bootstrap Standard Error

! Definition

The **bootstrap standard error** of a statistic $\hat{\theta}$ is the standard deviation of its bootstrap distribution:

$$SE_{boot}(\hat{\theta}) = \sqrt{\frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}_b^* - \bar{\hat{\theta}}^*)^2}$$

where $\bar{\hat{\theta}}^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_b^*$

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where $\bar{\hat{\theta}}^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_b^*$

This gives us an **empirical estimate** of the standard error without assuming any specific distribution!

Medical Example: Blood Pressure Reduction

A pilot study measures systolic blood pressure reduction (mmHg) after 8 weeks on a new antihypertensive:

```
1 bp_reduction <- c(12, 8, 15, 22, 5, 18, 11, 14, 9, 25, 7, 16, 13, 10, 19)
2 n_bp <- length(bp_reduction)
3
4 cat("Sample size:", n_bp, "\n")
```

Sample size: 15

```
1 cat("Mean reduction:", round(mean(bp_reduction), 2), "mmHg\n")
```

Mean reduction: 13.6 mmHg

```
1 cat("Sample SD:", round(sd(bp_reduction), 2), "mmHg\n")
```

Sample SD: 5.67 mmHg

```
1 cat("Classical SE:", round(sd(bp_reduction)/sqrt(n_bp), 2), "mmHg")
```

Classical SE: 1.46 mmHg

Bootstrap SE for Blood Pressure

```
1 set.seed(551)
2 B <- 10000
3 boot_means_bp <- replicate(B, {
4   boot_sample <- sample(bp_reduction, n_bp, replace = TRUE)
5   mean(boot_sample)
6 })
7
8 # Compare standard errors
9 classical_se <- sd(bp_reduction) / sqrt(n_bp)
10 boot_se <- sd(boot_means_bp)
11
12 cat("Classical SE (s/âˆšn):", round(classical_se, 3), "\n")
```

Classical SE (s/âˆšn): 1.463

```
1 cat("Bootstrap SE:", round(boot_se, 3), "\n")
```

Bootstrap SE: 1.41

```
1 cat("Ratio:", round(boot_se / classical_se, 3))
```

Ratio: 0.964

Bootstrap SE for Blood Pressure

```
1 set.seed(551)
2 B <- 10000
3 boot_means_bp <- replicate(B, {
4   boot_sample <- sample(bp_reduction, n_bp, replace = TRUE)
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6 })
7
8 # Compare standard errors
9 classical_se <- sd(bp_reduction) / sqrt(n_bp)
10 boot_se <- sd(boot_means_bp)
11
12 cat("Classical SE ( $s/\sqrt{n}$ ):", round(classical_se, 3), "\n")
```

Classical SE ($s/\sqrt{n}$): 1.463

```
1 cat("Bootstrap SE:", round(boot_se, 3), "\n")
```

Bootstrap SE: 1.41

```
1 cat("Ratio:", round(boot_se / classical_se, 3))
```

Ratio: 0.964

The bootstrap SE is very close to the classical SE when assumptions are met!

When Bootstrap SE Shines: Trimmed Mean

For **trimmed means** (robust to outliers), we don't have simple SE formulas.

```
1 # Data with potential outlier
2 response_times <- c(8, 12, 15, 7, 22, 45, 11, 9, 14, 18, 10, 13)
3
4 # Bootstrap for 10% trimmed mean
5 set.seed(551)
6 boot_trimmed <- replicate(10000, {
7   x <- sample(response_times, length(response_times), replace = TRUE)
8   mean(x, trim = 0.10) # Trim 10% from each end
9 })
10
11 cat("Sample mean:", round(mean(response_times), 2), "\n")
```

Sample mean: 15.33

```
1 cat("10% Trimmed mean:", round(mean(response_times, trim = 0.10), 2), "\n")
```

10% Trimmed mean: 13.2

```
1 cat("Bootstrap SE of trimmed mean:", round(sd(boot_trimmed), 2))
```

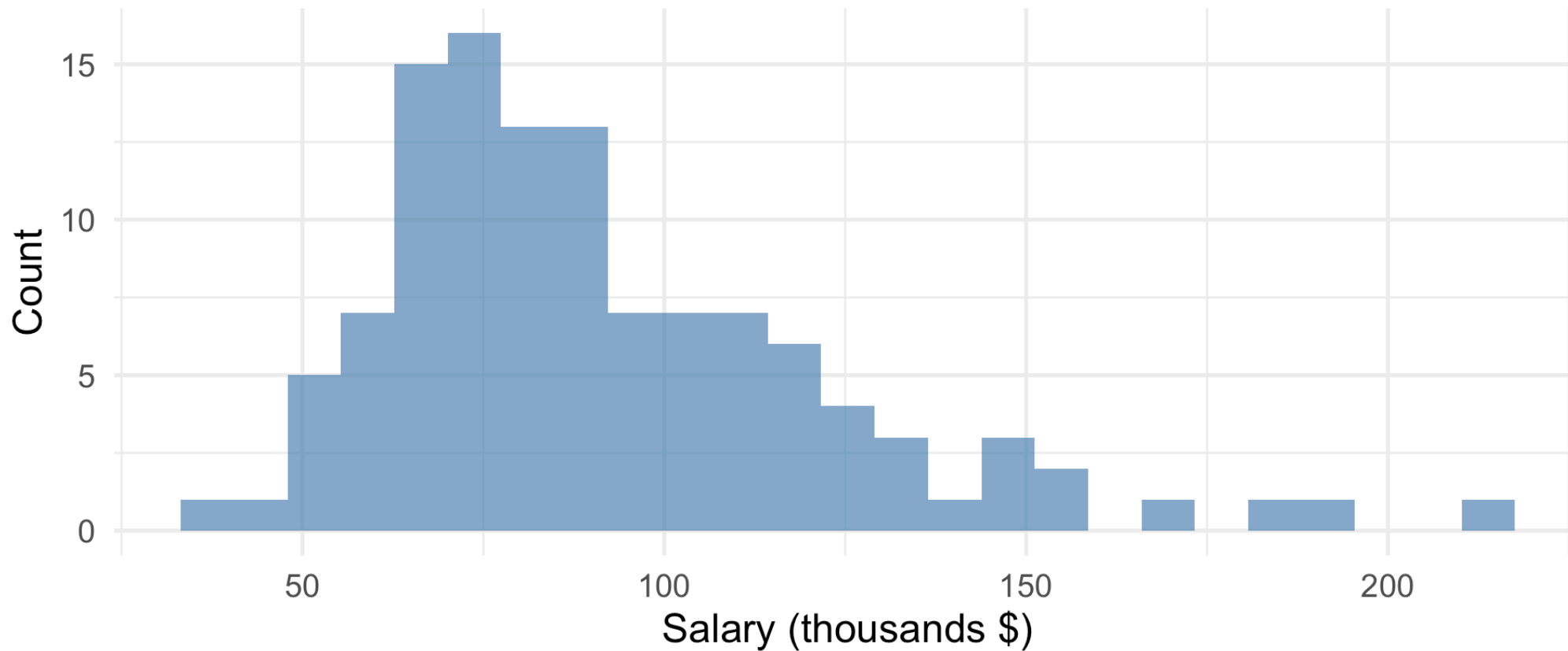
Bootstrap SE of trimmed mean: 2.67

Example: Database Administrator Salaries (Devore 8.18)

► Code

Distribution of Database Administrator Salaries

n = 115, with right skew



Bootstrap for Trimmed Mean Salary

```
1 # 10% trimmed mean
2 xbar_tr <- mean(salaries, trim = 0.10)
3
4 # Bootstrap
5 set.seed(551)
6 B <- 10000
7 boot_trimmed_sal <- replicate(B, {
8   x <- sample(salaries, n_sal, replace = TRUE)
9   mean(x, trim = 0.10)
10 })
11
12 cat("10% Trimmed mean:", round(xbar_tr, 0), "\n")
```

10% Trimmed mean: 88430

```
1 cat("Bootstrap SE:", round(sd(boot_trimmed_sal), 0), "\n")
```

Bootstrap SE: 2808

```
1 cat("Bootstrap mean of trimmed means:", round(mean(boot_trimmed_sal), 0))
```

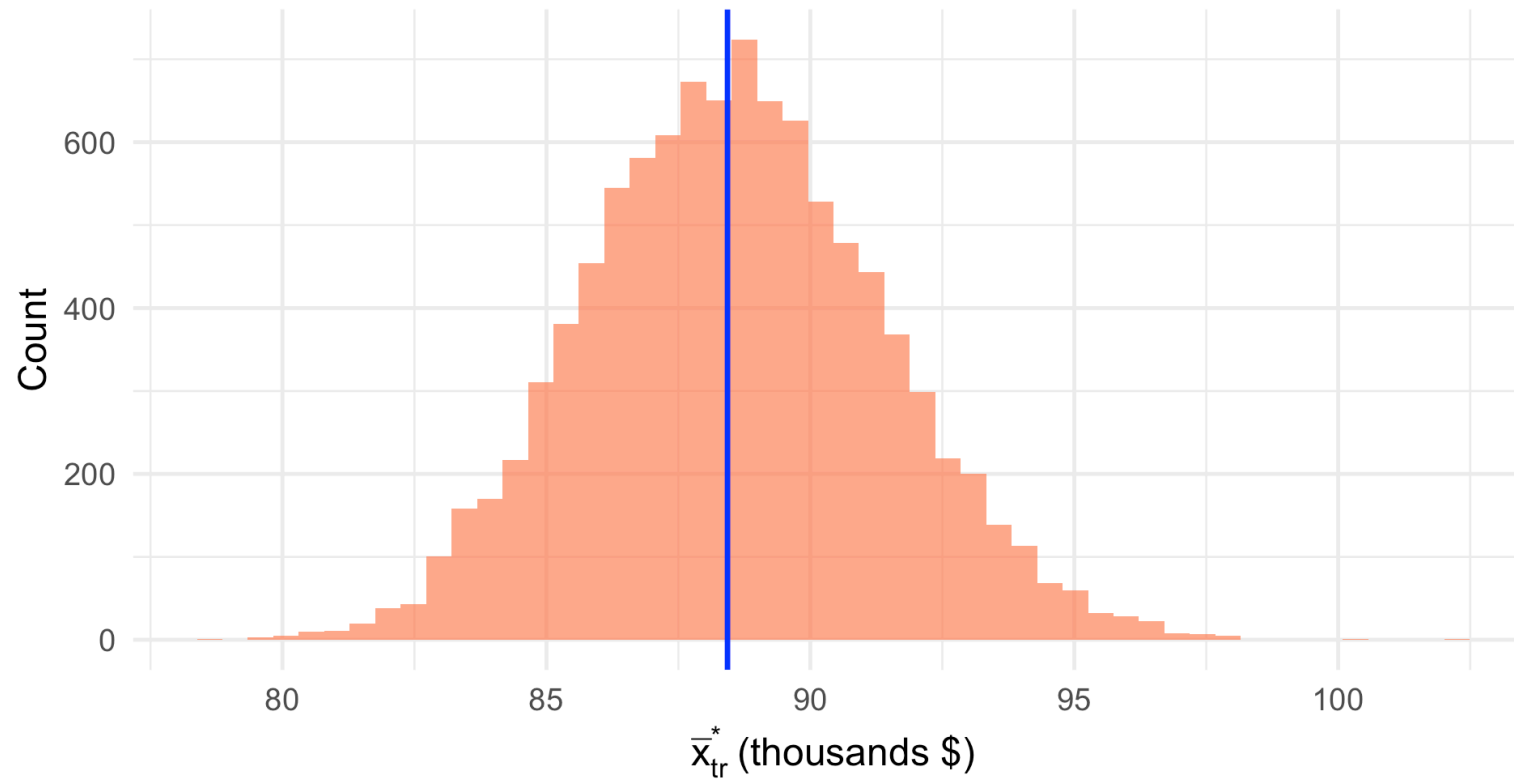
Bootstrap mean of trimmed means: 88561

Visualizing the Bootstrap Distribution

► Code

Bootstrap Distribution of 10% Trimmed Mean Salary

SE = \$2.8k



The bootstrap distribution appears approximately normal – good for t-based intervals!

Bias in Bootstrap Distributions (Chihara 5.6)

i Bootstrap Bias Estimate

The **bootstrap estimate of bias** is:

$$\widehat{\text{Bias}} = \bar{\hat{\theta}}^* - \hat{\theta}$$

where $\bar{\hat{\theta}}^*$ is the mean of the bootstrap distribution and $\hat{\theta}$ is the original sample statistic.

```
1 # Check bias for trimmed mean
2 boot_bias <- mean(boot_trimmed_sal) - xbar_tr
3 cat("Bootstrap bias estimate:", round(boot_bias, 0), "\n")
```

Bootstrap bias estimate: 131

```
1 cat("Bias as % of estimate:", round(100 * boot_bias / xbar_tr, 2), "%")
```

Bias as % of estimate: 0.15 %

Bias in Bootstrap Distributions (Chihara 5.6)

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```
1 # Check bias for trimmed mean
2 boot_bias <- mean(boot_trimmed_sal) - xbar_tr
3 cat("Bootstrap bias estimate:", round(boot_bias, 0), "\n")
```

Bootstrap bias estimate: 131

```
1 cat("Bias as % of estimate:", round(100 * boot_bias / xbar_tr, 2), "%")
```

Bias as % of estimate: 0.15 %

Small bias suggests the bootstrap distribution is well-centered!

Bootstrap Confidence Intervals

Types of Bootstrap Confidence Intervals

Several methods exist for constructing bootstrap CIs:

1. **Bootstrap Percentile Interval** â€” Simple, intuitive
2. **Bootstrap t Interval** â€” Better theoretical properties
3. **BCa Interval** â€” Bias-corrected and accelerated (advanced)

Types of Bootstrap Confidence Intervals

Several methods exist for constructing bootstrap CIs:

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Practical Guidance

- **Percentile interval:** Good when bootstrap distribution is symmetric, low bias
- **Bootstrap t interval:** Better for small samples, handles skewness
- Both require the bootstrap distribution to be a reasonable approximation

Bootstrap Percentile Interval (Devore 8.5)

! Definition

A **95% bootstrap percentile interval** for parameter θ has endpoints equal to the **2.5th and 97.5th percentiles** of the bootstrap distribution.

More generally, a $100(1 - \alpha)\%$ bootstrap percentile interval uses the $\alpha/2$ and $1 - \alpha/2$ quantiles:

$$\left(\hat{\theta}_{\alpha/2}^*, \hat{\theta}_{1-\alpha/2}^* \right)$$

Bootstrap Percentile Interval (Devore 8.5)

! Definition

A **95% bootstrap percentile interval** for parameter θ has endpoints equal to the **2.5th and 97.5th percentiles** of the bootstrap distribution.

More generally, a $100(1 - \alpha)\%$ bootstrap percentile interval uses the $\alpha/2$ and $1 - \alpha/2$ quantiles:

$$\left(\hat{\theta}_{\alpha/2}^*, \hat{\theta}_{1-\alpha/2}^* \right)$$

Intuition: If 95% of bootstrap statistics fall between L and U , then (L, U) is a plausible range for the true parameter.

Computing Percentile Intervals in R

```
1 # Blood pressure reduction data
2 set.seed(551)
3 bp_reduction <- c(12, 8, 15, 22, 5, 18, 11, 14, 9, 25, 7, 16, 13, 10, 19)
4
5 # Bootstrap
6 boot_means_bp <- replicate(10000, mean(sample(bp_reduction, length(bp_reduction), replace =
7
8 # 95% Percentile interval
9 percentile_ci <- quantile(boot_means_bp, probs = c(0.025, 0.975))
10
11 cat("Sample mean:", round(mean(bp_reduction), 2), "mmHg\n")
```

Sample mean: 13.6 mmHg

```
1 cat("95% Bootstrap Percentile CI:", round(percentile_ci, 2), "mmHg\n")
```

95% Bootstrap Percentile CI: 10.93 16.47 mmHg

```
1 # Compare to t-interval
2 t_ci <- t.test(bp_reduction)$conf.int
3 cat("95% t-interval:", round(t_ci, 2), "mmHg")
```

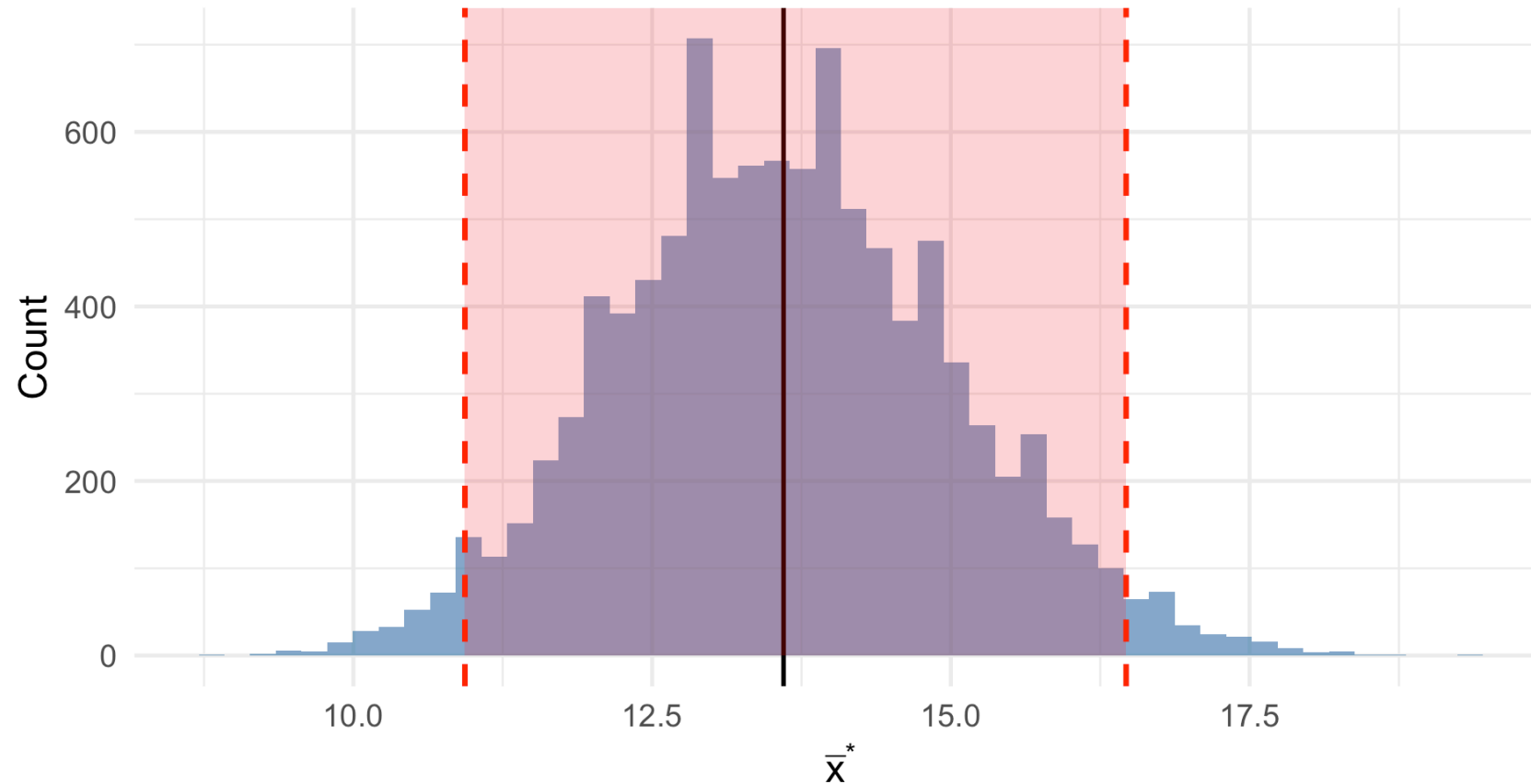
95% t-interval: 10.46 16.74 mmHg

Visualizing the Percentile Interval

► Code

Bootstrap Distribution with 95% Percentile Interval

CI: (10.9, 16.5) mmHg



Medical Example: Median Survival Time

For **median** survival, we don't have simple formulas for CIs. Bootstrap is ideal!

```
1 # Survival times (months) for 20 cancer patients
2 survival_times <- c(4, 8, 12, 15, 18, 6, 9, 24, 36, 11,
3                    7, 14, 21, 10, 13, 5, 17, 22, 8, 19)
4
5 # Bootstrap for median
6 set.seed(551)
7 boot_medians <- replicate(10000, {
8   median(sample(survival_times, length(survival_times), replace = TRUE))
9 })
10
11 # Results
12 cat("Sample median:", median(survival_times), "months\n")
```

Sample median: 12.5 months

```
1 cat("Bootstrap SE:", round(sd(boot_medians), 2), "months\n")
```

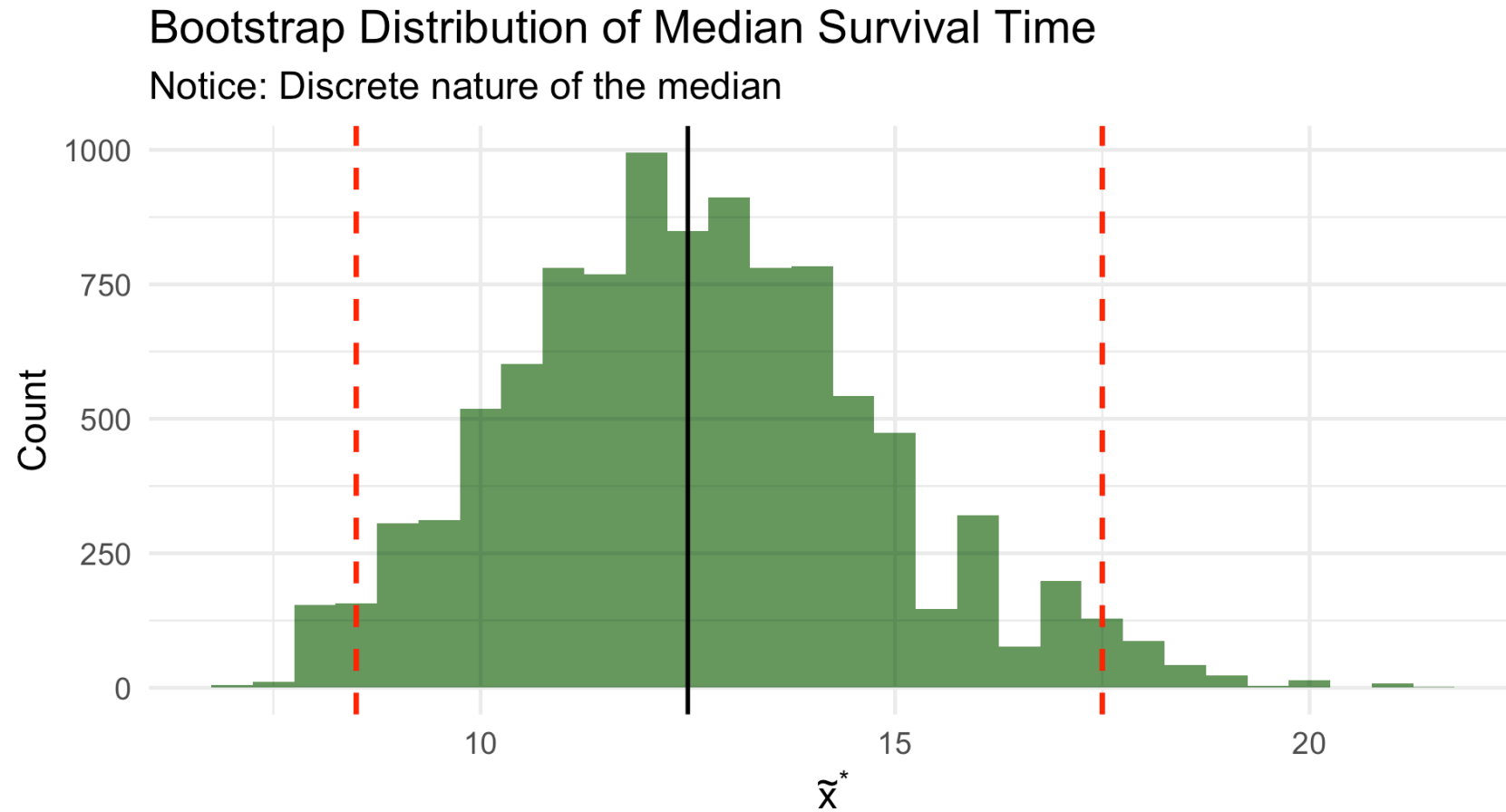
Bootstrap SE: 2.22 months

```
1 cat("95% Percentile CI:", quantile(boot_medians, c(0.025, 0.975)), "months")
```

95% Percentile CI: 8.5 17.5 months

Bootstrap Distribution of Median

► Code



The bootstrap median distribution takes on **few unique values** â€” this is typical!

Bootstrap t Confidence Interval (Devore 8.5)

! Bootstrap t Interval (Simple Version)

If the bootstrap distribution is approximately **normal** with small **bias**, use:

$$\hat{\theta} \pm t_{\alpha/2, n-1} \cdot SE_{boot}$$

where SE_{boot} is the bootstrap standard error.

Bootstrap t Confidence Interval (Devore 8.5)

! Bootstrap t Interval (Simple Version)

If the bootstrap distribution is approximately **normal** with small **bias**, use:

$$\hat{\theta} \pm t_{\alpha/2, n-1} \cdot SE_{boot}$$

where SE_{boot} is the bootstrap standard error.

```
1 # Bootstrap t interval for trimmed mean salary
2 xbar_tr <- mean(salaries, trim = 0.10)
3 se_boot <- sd(boot_trimmed_sal)
4 t_crit <- qt(0.975, df = length(salaries) - 1)
5
6 ci_lower <- xbar_tr - t_crit * se_boot
7 ci_upper <- xbar_tr + t_crit * se_boot
8
9 cat("Bootstrap t 95% CI for trimmed mean salary:\n")
```

Bootstrap t 95% CI for trimmed mean salary:

```
1 cat("($", round(ci_lower, 0), ", $", round(ci_upper, 0), ")", sep = " ")
($82868, $93992)
```

The Studentized Bootstrap t Interval (Chihara 7.5)

! Bootstrap t Interval (More Sophisticated)

For each bootstrap sample, compute:

$$T^* = \frac{\bar{X}^* - \bar{x}}{S^*/\sqrt{n}}$$

Use quantiles $Q_{\alpha/2}^*$ and $Q_{1-\alpha/2}^*$ of the bootstrap t distribution:

$$\left(\bar{x} - Q_{1-\alpha/2}^* \cdot \frac{s}{\sqrt{n}}, \bar{x} - Q_{\alpha/2}^* \cdot \frac{s}{\sqrt{n}} \right)$$

The Studentized Bootstrap t Interval (Chihara 7.5)

! Bootstrap t Interval (More Sophisticated)

For each bootstrap sample, compute:

$$T^* = \frac{\bar{X}^* - \bar{x}}{S^*/\sqrt{n}}$$

Use quantiles $Q_{\alpha/2}^*$ and $Q_{1-\alpha/2}^*$ of the bootstrap t distribution:

$$\left(\bar{x} - Q_{1-\alpha/2}^* \cdot \frac{s}{\sqrt{n}}, \bar{x} - Q_{\alpha/2}^* \cdot \frac{s}{\sqrt{n}} \right)$$

This method adapts to **skewness** in the data!

Bootstrap t Interval: R Implementation

```
1 bootstrap_t_interval <- function(data, B = 10000, conf = 0.95) {
2   n <- length(data)
3   xbar <- mean(data)
4   se <- sd(data) / sqrt(n)
5
6   # Compute bootstrap t statistics
7   t_star <- replicate(B, {
8     boot_samp <- sample(data, n, replace = TRUE)
9     (mean(boot_samp) - xbar) / (sd(boot_samp) / sqrt(n))
10  })
11
12  # Quantiles
13  alpha <- 1 - conf
14  q_lower <- quantile(t_star, alpha/2)
15  q_upper <- quantile(t_star, 1 - alpha/2)
16
17  # CI (note the reversal!)
18  c(xbar - q_upper * se, xbar - q_lower * se)
19 }
20
21 # Apply to blood pressure data
22 boot_t_ci <- bootstrap_t_interval(bp_reduction)
23 cat("Bootstrap t CI: ", round(boot_t_ci, 2), "mmHg")
```

Why the Reversal in Bootstrap t?

Understanding the Bootstrap t Formula

The percentile interval uses quantiles **directly**: $(\hat{\theta}_{\alpha/2}^*, \hat{\theta}_{1-\alpha/2}^*)$

The bootstrap t interval uses quantiles of the **t-statistics** with reversal:

$$\left(\bar{x} - Q_{1-\alpha/2}^* \cdot SE, \bar{x} - Q_{\alpha/2}^* \cdot SE \right)$$

The reversal accounts for the fact that if T^* is large, then $\bar{X}^*$ was above $\bar{x}$, so the lower bound for μ should be reduced.

Why the Reversal in Bootstrap t?

Understanding the Bootstrap t Formula

The percentile interval uses quantiles **directly**: $(\hat{\theta}_{\alpha/2}^*, \hat{\theta}_{1-\alpha/2}^*)$

The bootstrap t interval uses quantiles of the **t-statistics** with reversal:

$$\left(\bar{x} - Q_{1-\alpha/2}^* \cdot SE, \bar{x} - Q_{\alpha/2}^* \cdot SE \right)$$

The reversal accounts for the fact that if T^* is large, then $\bar{X}^*$ was above $\bar{x}$, so the lower bound for μ should be reduced.

This is analogous to how the classical t-interval works: large t values suggest the true mean might be lower than $\bar{x}$.

Comparing All Methods

► Code

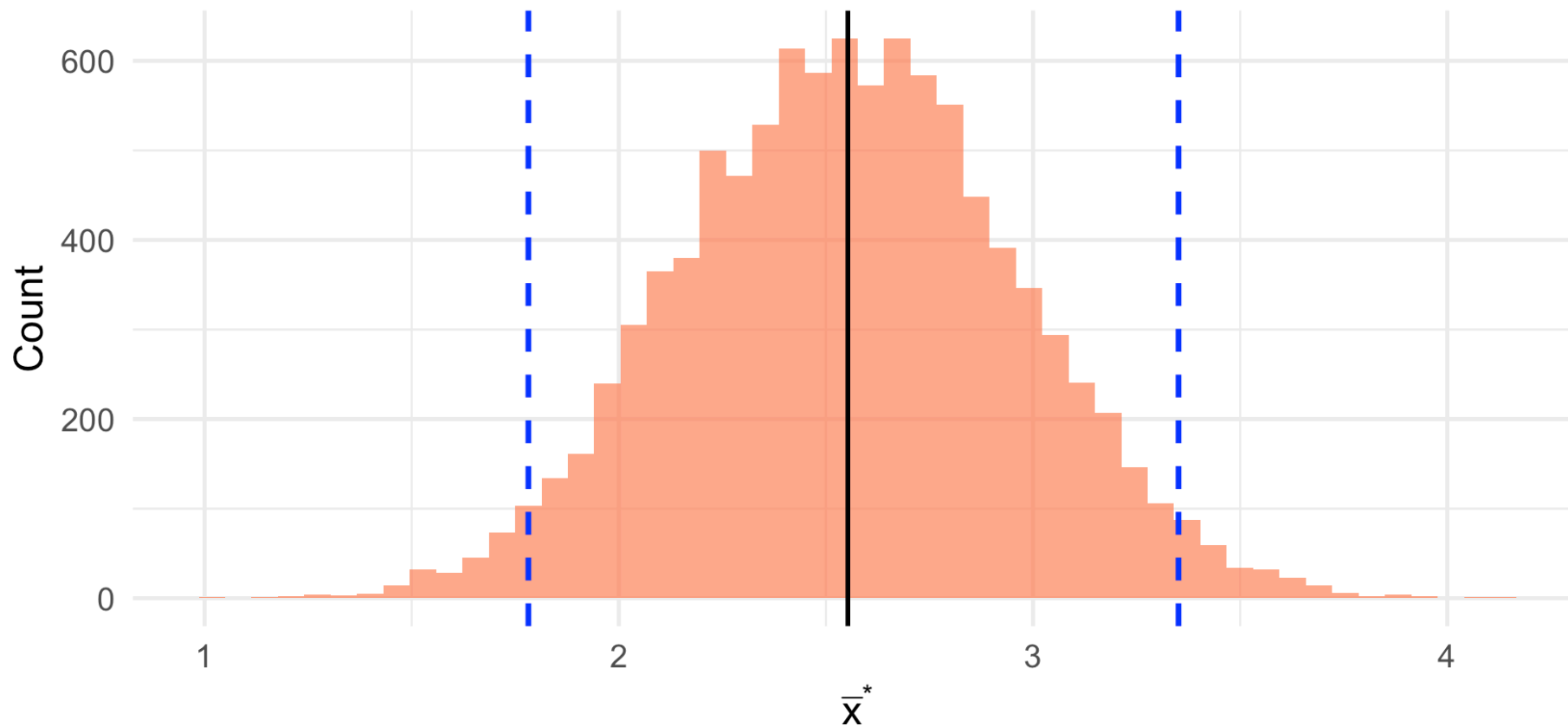
```
# A tibble: 3 × 4
  Method      Lower Upper Width
  <chr>      <dbl> <dbl> <dbl>
1 Classical t    10.5  16.7  6.28
2 Bootstrap Percentile 10.9  16.5  5.53
3 Bootstrap t    10.8  17.2  6.47
```

When Percentile May Be Suboptimal: Skewed Distributions

► Code

Bootstrap Distribution from Skewed Population

Percentile interval may have incorrect coverage



Practical Considerations

How Many Bootstrap Samples? (Chihara 5.9)

Guidelines

- **Standard errors:** $B = 1,000$ is often sufficient
- **Confidence intervals:** $B = 5,000$ to $10,000$ recommended
- **Percentile tails:** More samples needed for accurate tail quantiles

```
1 # Demonstrating convergence of bootstrap SE
2 set.seed(551)
3 B_values <- c(100, 500, 1000, 5000, 10000)
4
5 convergence <- map_dfr(B_values, function(B) {
6   boot_means <- replicate(B, mean(sample(bp_reduction, length(bp_reduction), replace = TRUE)
7   tibble(B = B, SE = sd(boot_means))
8 })
9
10 convergence
```

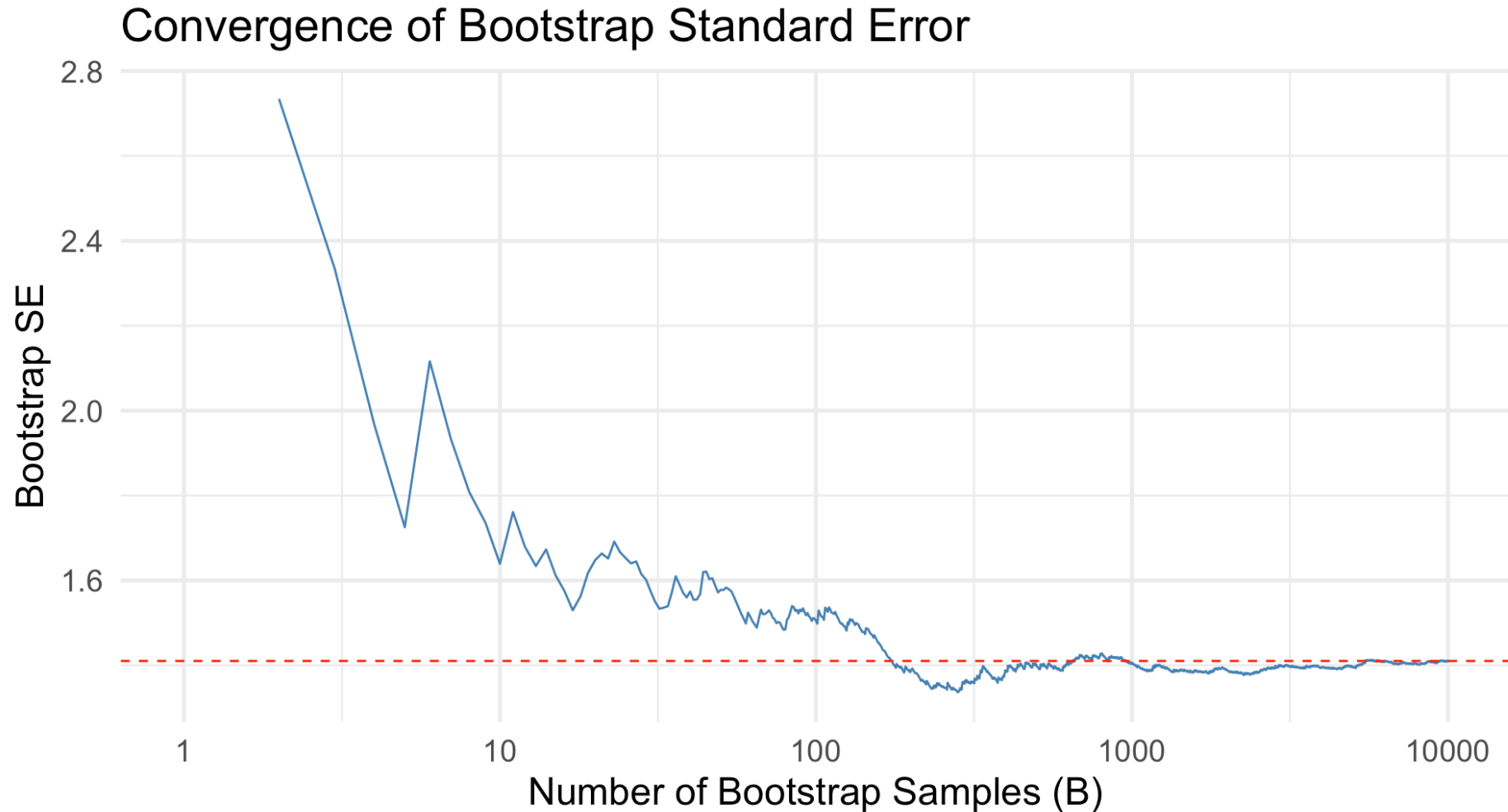
A tibble: 5 × 2

	B	SE
	<dbl>	<dbl>
1	100	1.51
2	500	1.37
3	1000	1.38
4	5000	1.41



Visualizing Bootstrap Convergence

► Code



Classical vs. Bootstrap: Guidelines (Chihara 7.5.1)

Situation	Recommendation
Small n ($\hat{\sigma} \approx 10$), normal data	Classical t preferred
Small n ($\hat{\sigma} \approx 10$), skewed data	Bootstrap t (with caution)
Large n , normal data	Either works well
Large n , skewed data	Bootstrap t preferred
Complex statistic (trimmed mean, ratio)	Bootstrap (no classical alternative)

Classical vs. Bootstrap: Guidelines (Chihara 7.5.1)

Situation	Recommendation
Small n ($\hat{\sigma} \approx 10$), normal data	Classical t preferred
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Large n , normal data	Either works well
Large n , skewed data	Bootstrap t preferred
Complex statistic (trimmed mean, ratio)	Bootstrap (no classical alternative)

Key insight: The bootstrap **learns skewness from the sample**, but small samples may not accurately represent population skewness.

When Does Bootstrap Work Well?

Bootstrap Works Well When:

1. The sample is **representative** of the population
2. The sample size is **not too small** ($n \geq 15$ typically)
3. The statistic is **smooth** (means, variances, regression coefficients)
4. The parameter is well-defined and has finite variance

When Does Bootstrap Work Well?

Bootstrap Works Well When:

1. The sample is **representative** of the population
2. The sample size is **not too small** ($n \geq 15$ typically)
3. The statistic is **smooth** (means, variances, regression coefficients)
4. The parameter is well-defined and has finite variance

Bootstrap May Struggle When:

1. Sample is very small ($n < 10$)
2. Data has extreme outliers not representative of population
3. Statistic is very discrete (e.g., median with few unique values)
4. Estimating extreme quantiles
5. Population has infinite variance (Cauchy, Pareto with $\alpha \leq 2$)

Medical Applications

Medical Application: Relative Risk (Chihara 5.5)

In a cardiovascular study:

- High BP group: 55 events in 3,338 men
- Low BP group: 21 events in 2,676 men

```
1 # Relative risk
2 p1 <- 55/3338 # High BP group
3 p2 <- 21/2676 # Low BP group
4 rr_hat <- p1/p2
5
6 cat("Risk (high BP):", round(p1 * 100, 2), "%\n")
```

Risk (high BP): 1.65 %

```
1 cat("Risk (low BP):", round(p2 * 100, 2), "%\n")
```

Risk (low BP): 0.78 %

```
1 cat("Relative Risk:", round(rr_hat, 2))
```

Relative Risk: 2.1

Bootstrap for Relative Risk

```
1 set.seed(551)
2 n1 <- 3338; x1 <- 55
3 n2 <- 2676; x2 <- 21
4
5 # Bootstrap
6 boot_rr <- replicate(10000, {
7   # Resample from each group (binary outcomes)
8   boot_p1 <- sum(sample(c(rep(1, x1), rep(0, n1 - x1)), n1, replace = TRUE)) / n1
9   boot_p2 <- sum(sample(c(rep(1, x2), rep(0, n2 - x2)), n2, replace = TRUE)) / n2
10
11   # Handle division by zero
12   if (boot_p2 == 0) return(NA)
13   boot_p1 / boot_p2
14 })
15
16 boot_rr <- boot_rr[!is.na(boot_rr)]
17
18 cat("Bootstrap SE:", round(sd(boot_rr), 3), "\n")
```

Bootstrap SE: 0.627

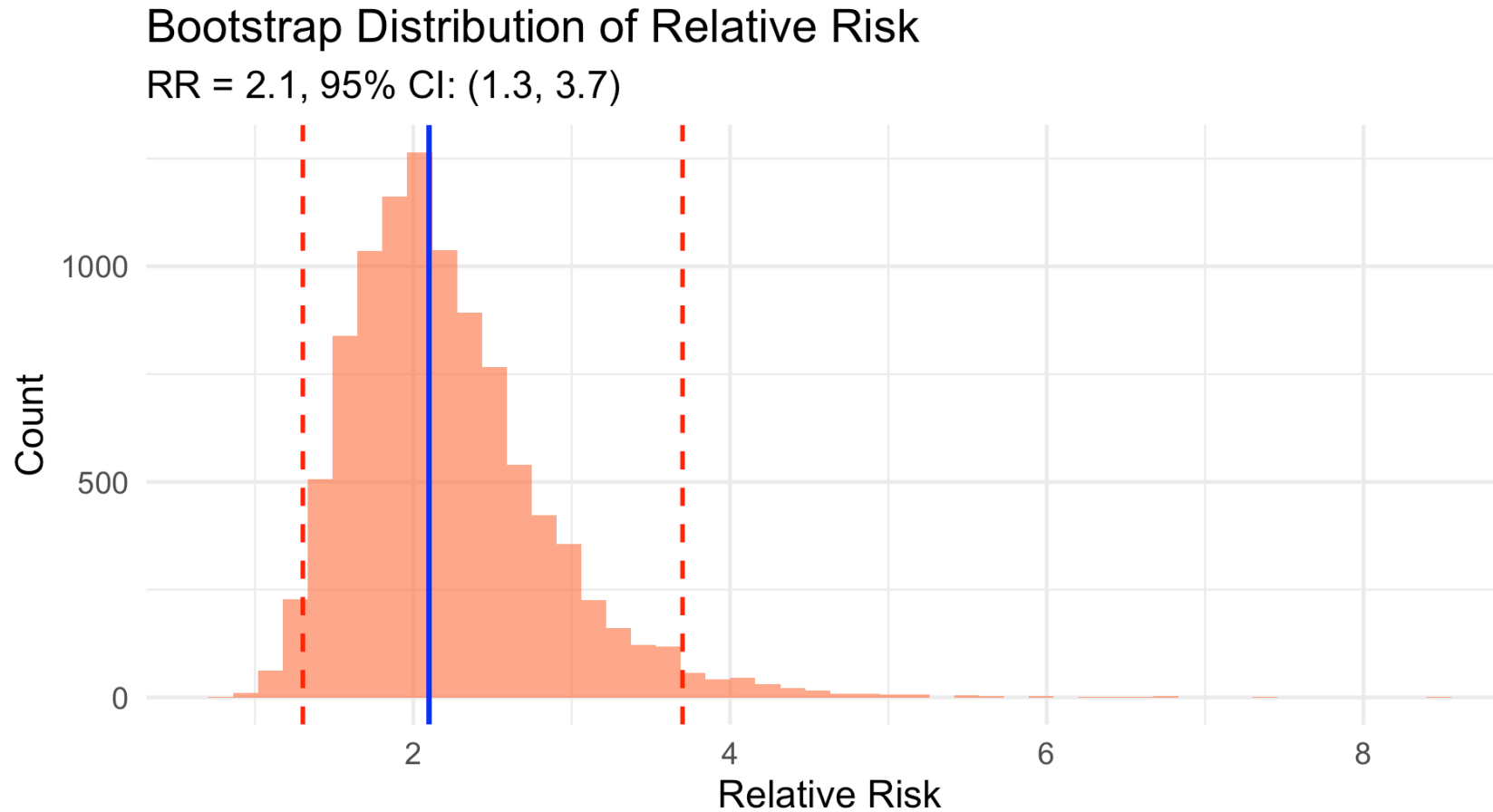
```
1 cat("95% Percentile CI:", round(quantile(boot_rr, c(0.025, 0.975)), 2))
```

95% Percentile CI: 1.3 3.7



Relative Risk: Bootstrap Distribution

► Code



Interpretation: High BP patients have 1.3 to 3.7 times the cardiovascular risk (95% CI).

Clinical Trial: Comparing Treatments

A trial compares time to pain relief for two analgesics:

```
1 set.seed(551)
2 # Treatment A: faster, less variable
3 drug_A <- c(12, 18, 15, 22, 14, 16, 20, 13, 17, 19)
4 # Treatment B: slower, more variable
5 drug_B <- c(25, 18, 35, 22, 28, 15, 42, 20, 30, 24)
6
7 cat("Drug A – Mean:", round(mean(drug_A), 1), ", SD:", round(sd(drug_A), 1), "\n")
```

Drug A – Mean: 16.6 , SD: 3.2

```
1 cat("Drug B – Mean:", round(mean(drug_B), 1), ", SD:", round(sd(drug_B), 1), "\n")
```

Drug B – Mean: 25.9 , SD: 8.2

```
1 cat("Difference (B – A):", round(mean(drug_B) – mean(drug_A), 1), "minutes")
```

Difference (B – A): 9.3 minutes

Two-Sample Bootstrap

```
1 set.seed(551)
2 # Bootstrap the difference in means
3 boot_diff <- replicate(10000, {
4   boot_A <- sample(drug_A, length(drug_A), replace = TRUE)
5   boot_B <- sample(drug_B, length(drug_B), replace = TRUE)
6   mean(boot_B) - mean(boot_A)
7 })
8
9 # Results
10 diff_ci <- quantile(boot_diff, c(0.025, 0.975))
11 cat("Bootstrap SE:", round(sd(boot_diff), 2), "\n")
```

Bootstrap SE: 2.62

```
1 cat("95% CI for difference:", round(diff_ci, 2), "minutes")
```

95% CI for difference: 4.5 14.7 minutes

Two-Sample Bootstrap

```
1 set.seed(551)
2 # Bootstrap the difference in means
3 boot_diff <- replicate(10000, {
4   boot_A <- sample(drug_A, length(drug_A), replace = TRUE)
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6   mean(boot_B) - mean(boot_A)
7 })
8
9 # Results
10 diff_ci <- quantile(boot_diff, c(0.025, 0.975))
11 cat("Bootstrap SE:", round(sd(boot_diff), 2), "\n")
```

Bootstrap SE: 2.62

```
1 cat("95% CI for difference:", round(diff_ci, 2), "minutes")
```

95% CI for difference: 4.5 14.7 minutes

Since CI excludes 0, there's evidence Drug B takes longer to provide relief.

Bootstrap for Odds Ratio

In a case-control study of lung cancer and smoking:

```
1 # Lung cancer cases
2 cases_smokers <- 85; cases_nonsmokers <- 15
3 # Controls
4 controls_smokers <- 60; controls_nonsmokers <- 40
5
6 # Sample odds ratio
7 or_hat <- (cases_smokers * controls_nonsmokers) / (cases_nonsmokers * controls_smokers)
8 cat("Sample Odds Ratio:", round(or_hat, 2))
```

Sample Odds Ratio: 3.78

Bootstrap for Odds Ratio (cont.)

```
1 set.seed(551)
2 # Bootstrap
3 n_cases <- cases_smokers + cases_nonsmokers
4 n_controls <- controls_smokers + controls_nonsmokers
5
6 boot_or <- replicate(10000, {
7   # Resample cases
8   boot_cases <- sample(c(rep(1, cases_smokers), rep(0, cases_nonsmokers)),
9                        n_cases, replace = TRUE)
10  boot_case_smoke <- sum(boot_cases)
11  boot_case_nonsmoke <- n_cases - boot_case_smoke
12
13  # Resample controls
14  boot_controls <- sample(c(rep(1, controls_smokers), rep(0, controls_nonsmokers)),
15                        n_controls, replace = TRUE)
16  boot_ctrl_smoke <- sum(boot_controls)
17  boot_ctrl_nonsmoke <- n_controls - boot_ctrl_smoke
18
19  # OR (with continuity correction)
20  (boot_case_smoke + 0.5) * (boot_ctrl_nonsmoke + 0.5) /
21  ((boot_case_nonsmoke + 0.5) * (boot_ctrl_smoke + 0.5))
22 }
```



Comparing Bootstrap CI Methods

Review: Three Bootstrap CI Methods

We've now seen three approaches to bootstrap confidence intervals:

Method	Formula	Key Idea
Percentile	$(\hat{\theta}_{\alpha/2}^*, \hat{\theta}_{1-\alpha/2}^*)$	Use quantiles of bootstrap distribution directly
Simple Bootstrap t	$\hat{\theta} \pm t_{\alpha/2, n-1} \cdot SE_{boot}$	Bootstrap for SE, t-distribution for critical values
Studentized Bootstrap t	$(\bar{x} - Q_{1-\alpha/2}^* \cdot SE, \bar{x} - Q_{\alpha/2}^* \cdot SE)$	Bootstrap t-statistics, use their quantiles

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Key question: When should we use each method?

The Two “Bootstrap t” Methods

Important Distinction

There are actually **two different methods** both called “bootstrap t”:

1. Simple Bootstrap t (Devore 8.5):

$$\hat{\theta} \pm t_{\alpha/2, n-1} \cdot SE_{boot}$$

- Uses bootstrap **only** to estimate SE
- Uses standard t critical values (± 1.96 , ± 2.045 , etc.)
- **Assumes** the bootstrap distribution is approximately normal

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2. Studentized Bootstrap t (Chihara 7.5):

$$\left(\bar{x} - Q_{1-\alpha/2}^* \cdot SE, \bar{x} - Q_{\alpha/2}^* \cdot SE \right)$$

- Computes $T^* = \frac{\bar{X}^* - \bar{x}}{S^* / \sqrt{n}}$ for each resample
- Uses quantiles from the **bootstrap t distribution**

A Common Misconception

The Intuition Trap

“The percentile interval can be asymmetric, so it should handle skewed distributions better than bootstrap t.”

This reasoning is incorrect!

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The Intuition Trap

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This reasoning is incorrect!

The confusion arises from conflating two different questions:

1. **Can the interval be asymmetric?** → Yes, both methods can produce asymmetric intervals
2. **Does the interval correctly adjust for skewness?** → The studentized bootstrap t does this better

Why Studentized Bootstrap t Handles Skewness Better

From Chihara and Hesterberg (Section 7.5.1):

“The bootstrap percentile makes **less of a skewness correction** than does the bootstrap t.”

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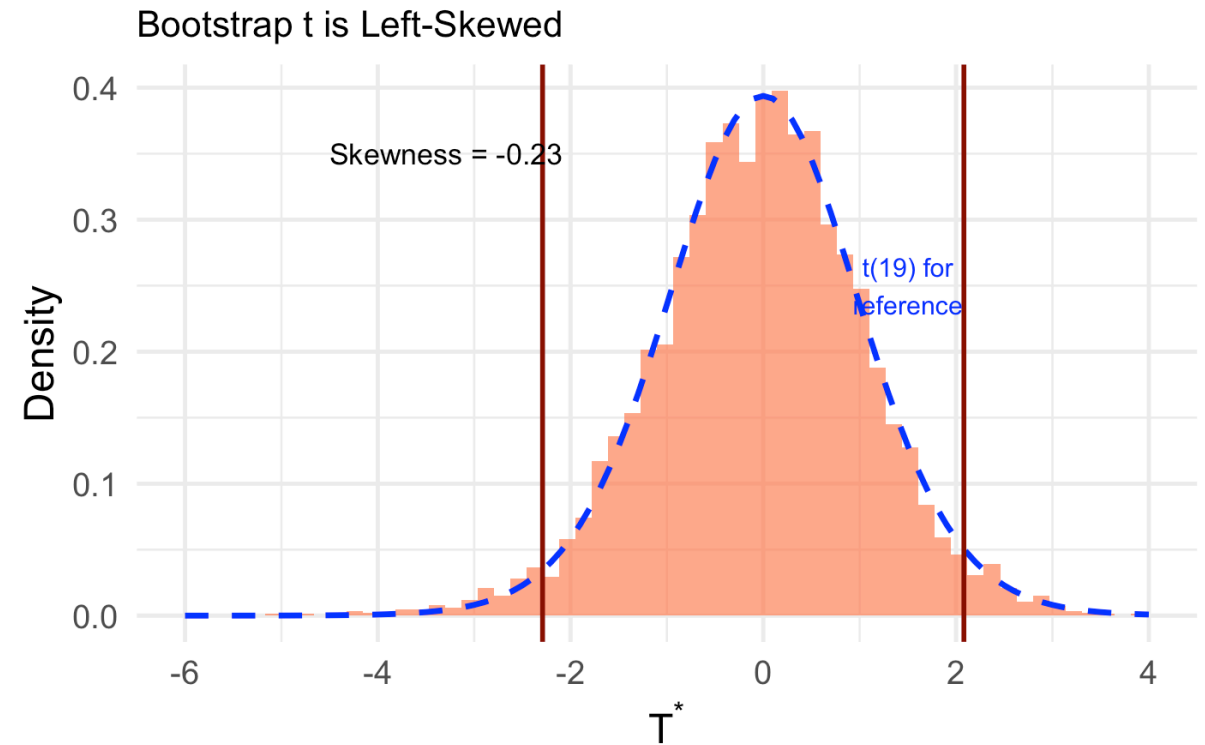
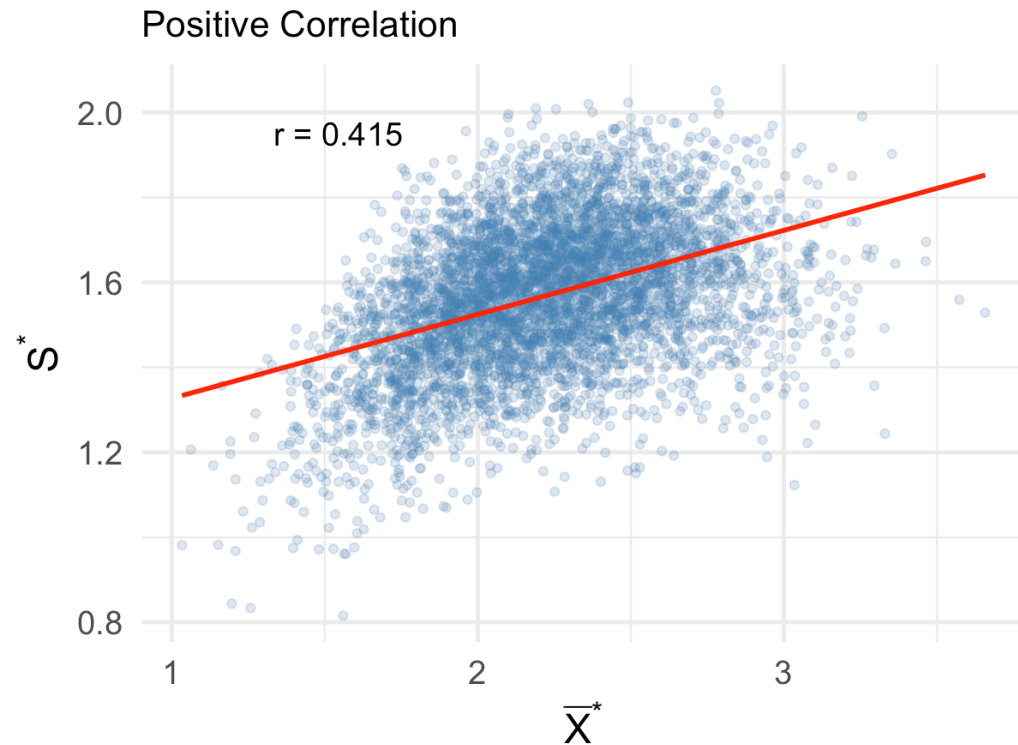
“The bootstrap percentile makes **less of a skewness correction** than does the bootstrap t.”

The key insight is **studentization** — dividing by $S^* / \sqrt{n}$ creates a correction:

- For **right-skewed data**: When $\bar{X}^* > \bar{x}$, typically $S^* > s$ also
- The larger denominator **dampens** extreme positive T^* values
- This creates left-skew in T^* that **counteracts** the coverage bias

Visualizing the Studentization Effect

► Code



The left tail of T^* extends further than the right — this asymmetry adjusts for the skewness in the data.

Theoretical Accuracy (Chihara 7.5.1)

! Coverage Error Rates

As sample size n increases by a factor of 10:

- **Studentized Bootstrap t:** Coverage error decreases by factor of **10**
- **Percentile:** Coverage error decreases by factor of $\sqrt{10} \approx 3.2$

Theoretical Accuracy (Chihara 7.5.1)

! Coverage Error Rates

As sample size n increases by a factor of 10:

- **Studentized Bootstrap t**: Coverage error decreases by factor of **10**
- **Percentile**: Coverage error decreases by factor of $\sqrt{10} \approx 3.2$

This means the studentized bootstrap t converges to correct coverage **much faster** than the percentile interval for skewed populations.

The percentile interval and classical t have similar (slower) convergence rates.

Simulation: Coverage Comparison

► Code

Coverage for Exp(0.5) data, $n = 15$ (1000 simulations)

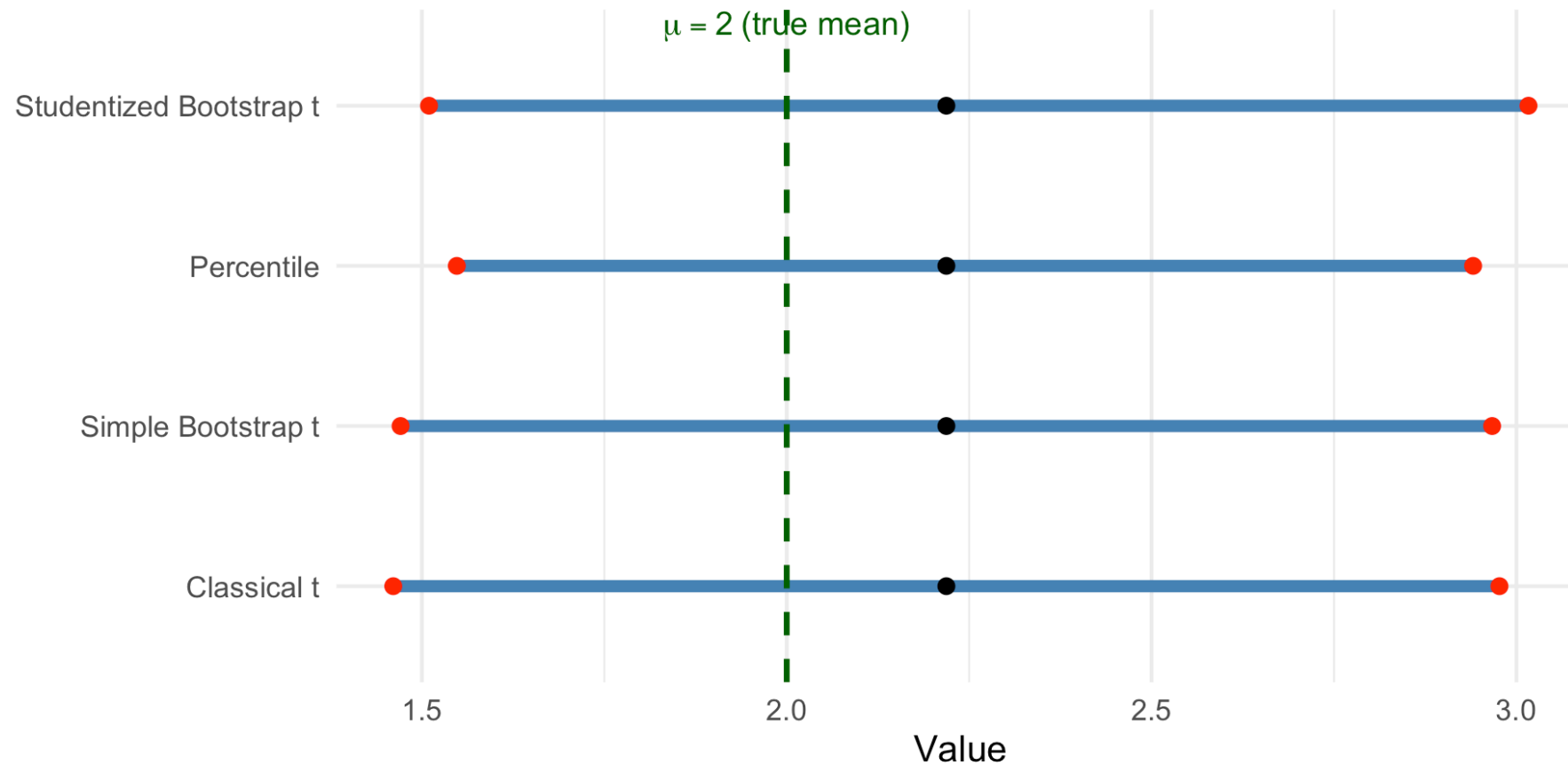
Method	Actual Coverage	Target
Classical t	92.9%	95%
Percentile	90.4%	95%
Studentized Bootstrap t	95.4%	95%

Visualizing the Intervals

► Code

Comparison of 95% CIs for Right-Skewed Data

Studentized bootstrap t extends further right to capture the true mean



Summary: Normality Assumptions

Method	Assumes Normal Population?	Assumes Normal Bootstrap Dist?	Handles Skewness?
Classical t	Yes (or large n for CLT)	—	Poorly
Simple Bootstrap t	No	Yes	Poorly
Studentized Bootstrap t	No	No	Well
Percentile	No	No	Moderately

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Percentile	No	No	Moderately

Key Insight

The **studentized bootstrap t** is the only method that adapts to skewness without assuming normality. It does this through studentization, not by assuming any particular distribution.

When to Use Each Method

Percentile Interval

- Bootstrap distribution is symmetric
- Low bias
- Simplicity is important
- SE is hard to compute (e.g., median)

Studentized Bootstrap t

- Data is skewed
- Accurate coverage is critical
- Sample size is moderate ($n \geq 15$)
- SE can be computed for each resample

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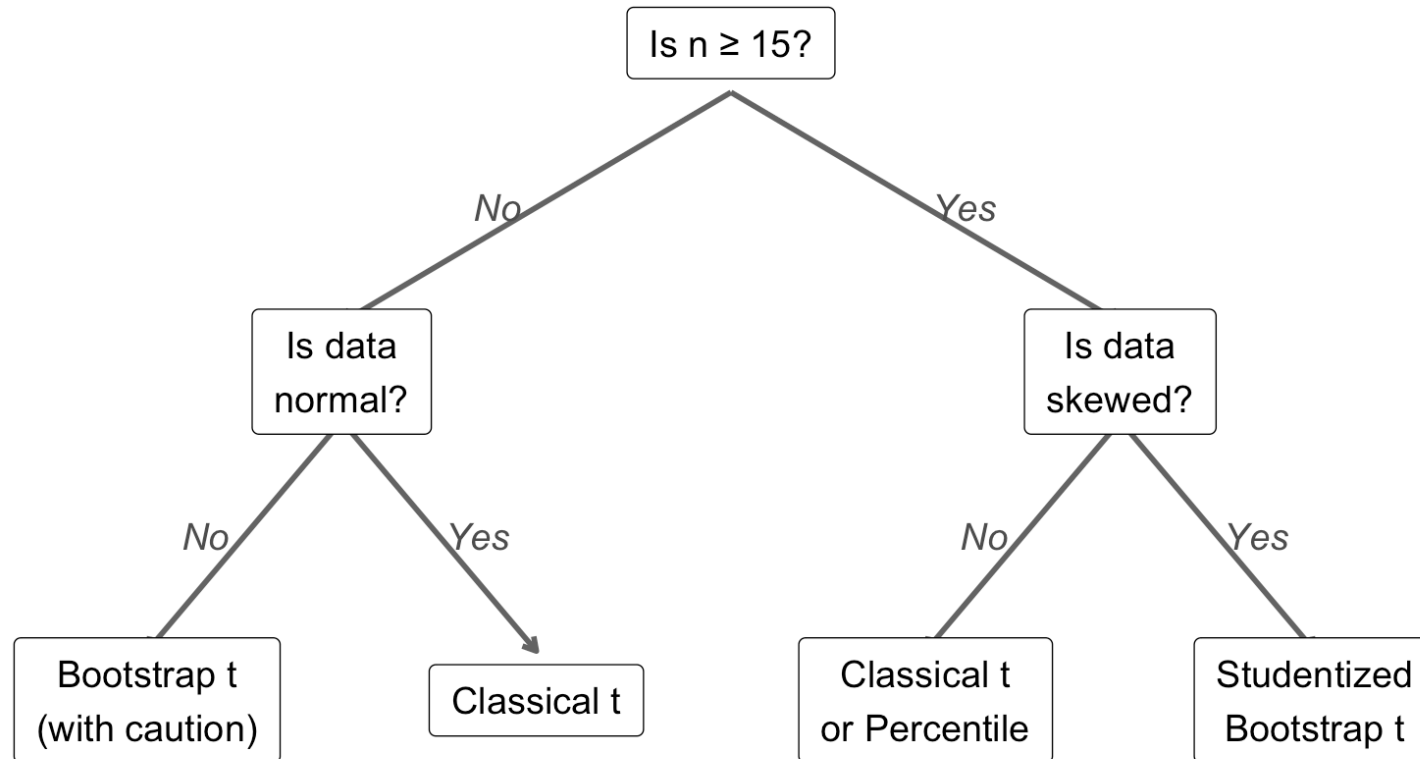
Small Sample Warning

For $n < 10$: **Classical t may be better!** The bootstrap cannot accurately estimate skewness from too few observations.

From Chihara & Hesterberg: “We recommend the formula t if $n \leq 10$, and the bootstrap t otherwise.”

Decision Guide

Choosing a Confidence Interval Method



Summary

Key Takeaways

! Bootstrap CI Selection

1. **Percentile interval:** Simple, intuitive, but provides less skewness correction
2. **Simple bootstrap t:** Uses bootstrap SE with t critical values; assumes approximate normality of bootstrap distribution
3. **Studentized bootstrap t:** Uses bootstrap t-statistics; adapts to skewness without assuming normality; best coverage for skewed data
4. **For small samples ($n < 10$):** Classical t may be preferable—bootstrap can't estimate skewness well
5. **BCa interval:** An advanced option that corrects for both bias and skewness (available in R packages)

What the Bootstrap Cannot Do

Bootstrap CAN estimate:

- Standard errors
- Bias
- Skewness of sampling distribution
- Confidence intervals

Bootstrap CANNOT:

- Improve point estimates
- Fix a bad/unrepresentative sample
- Create new information

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Remember

The bootstrap tells you about the **variability** of your estimate. It does not tell you whether your sample was good in the first place!

Looking Ahead

Next Topics:

- **Hypothesis testing framework** (Lesson 11)
- Type I and Type II errors
- P-values and their interpretation
- Power and sample size

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- Hypothesis testing framework (Lesson 11)
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Later:

- Two-sample inference
- Permutation tests
- Bootstrap hypothesis testing

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Section 8.5
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Chapters 5, 7
- Efron, B. and Tibshirani, R. (1993). *An Introduction to the Bootstrap*. Chapman & Hall.

Questions?

Thank you!